



# Ninaivu

நினைவு · memory

**For whoever looks after the technical side.**

Technical administrator guide · version 1.0.0

**Docker**  
on a NAS

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# Technical administrator guide

The guide tells a household how to install Ninaivu with the installer for their computer and use it. This page is for whoever looks after the technical side: the source code, Docker, running from a checkout, the command line, environment variables and where to report a problem. It is in English only.

## Where things are

|                                                    |                                                                                                                                         |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Source code                                        | <a href="https://github.com/javajaga-usa/Ninaivu">https://github.com/javajaga-usa/Ninaivu</a>                                           |
| Installers (the download page the guide points to) | <a href="https://github.com/javajaga-usa/Ninaivu/releases/latest">https://github.com/javajaga-usa/Ninaivu/releases/latest</a>           |
| Reporting a problem                                | <a href="https://github.com/javajaga-usa/Ninaivu/issues">https://github.com/javajaga-usa/Ninaivu/issues</a>                             |
| Extensions (Creative Studio, Gemini and the rest)  | <a href="https://github.com/javajaga-usa/Ninaivu/tree/main/extensions">https://github.com/javajaga-usa/Ninaivu/tree/main/extensions</a> |
| The docs site                                      | <a href="https://javajaga-usa.github.io/Ninaivu/">https://javajaga-usa.github.io/Ninaivu/</a>                                           |

When reporting a problem, say which version (**System** → **Settings** shows it under *This home*), what was done, and what the log said.

## More on the installers

- **Windows.** Once Ninaivu is in the winget catalogue, `winget install Ninaivu.Ninaivu` will work too.
- **macOS.** Once Ninaivu is in Homebrew, `brew install --cask ninaivu` will work too. `open -a Ninaivu --args --tray` opens the menu-bar tray instead of the Control Panel.
- **Linux and Raspberry Pi.** `--photos DIR`, `--prefix DIR`, `--no-service` and `--quiet` skip the installer's questions, for a headless Pi set up over SSH. `sudo sh Ninaivu-<version>-linux-arm64.sh` installs under `/opt/ninaivu` instead of `~/.local/share/ninaivu`. `~/.local/share/ninaivu/uninstall` removes it.

## Docker

For a NAS or a Linux box:

```
git clone https://github.com/javajaga-usa/Ninaivu.git
cd Ninaivu
cp installers/docker/.env.example installers/docker/.env # set MEDIA_DIR to your
photo folder
docker compose -f installers/docker/docker-compose.yml up -d
```

The family app answers on port 5000 to the whole network; the console on 3000, published on the NAS itself only ( 127.0.0.1:3000 ). To make the administrator from another computer, open the console through an SSH tunnel or on the NAS, and give the **setup code** printed in `docker logs ninaivu`. The `.env` file is only for folders and ports: everything about how Ninaivu behaves is chosen in the console, and nothing in `.env` undoes it at the next restart. The image leaves out search by description unless built with `--build-arg WITH_AI=1`. The operations guide covers systemd, reverse proxies, HTTPS and large libraries.

## From a checkout

Running from the source needs Python 3.12 or newer (the installers bring their own).

```
git clone https://github.com/javajaga-usa/Ninaivu.git
cd Ninaivu
start.cmd ~/Pictures # Windows: or double-click start.cmd
sh launcher/start.sh ~/Pictures # macOS and Linux
```

The launcher makes a virtual environment, installs what is missing (nothing large: search by description is added later, if you want it), serves HTTPS on port 443 — or 8080 on Linux, where an ordinary user may not use 443 — finds free ports and opens your browser. `python -m ninaivu <folder>` runs the server directly, on plain HTTP and port 80 (8080 without the right to use 80), unless given `--https`.

On Windows, `start.cmd` needs no Python beforehand: when it finds no Python 3.12 or newer, it installs Python 3.12 for the current user first (with winget, or from python.org when winget is missing), then carries on. No administrator rights are needed. **Start - Ninaivu Control Panel.vbs** opens the Control Panel once `start.cmd` has run.

On a Mac, double-click **Setup Ninaivu.command** in the Ninaivu folder once instead of using a terminal. It clears the download quarantine, installs Python 3.12 if the Mac has no Python 3.12 or newer (after asking: with Homebrew when the Mac has it, otherwise the installer from python.org, which is run only when it carries the Python Software Foundation's signature, and for which macOS asks your password), installs what Ninaivu needs, and puts two apps in Applications: **Ninaivu**, which starts the library in a Terminal window and asks once which folder holds your photos, and the **Ninaivu Control Panel**, which starts and stops it without one. `sh launcher/start.sh` offers the same Python install. More on the Control Panel.

A checkout keeps its log in `.ninaivu-control/server.log` beside it.

## Running it from the command line

`python -m ninaivu <folder>` (from a checkout) starts the server, and these are the flags worth knowing:

| Flag                                                              |                                                                                 |
|-------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <code>--admin USER:PASSWORD</code>                                | make the administrator without a browser                                        |
| <code>--port</code> , <code>--admin-port</code>                   | the two ports (80 — 443 with HTTPS — and 3000 by default)                       |
| <code>--local-only</code>                                         | this computer only; phones cannot reach it                                      |
| <code>--https</code> , <code>--cert</code> , <code>--key</code>   | HTTPS with Ninaivu's own certificate, or yours                                  |
| <code>--ai auto clip light off</code>                             | the image model; <i>auto</i> follows the hardware tier                          |
| <code>--faces</code> , <code>--ocr</code> , <code>--places</code> | turn the passes on from the start                                               |
| <code>--no-watch</code>                                           | do not watch the folders for changes                                            |
| <code>--open-browsing</code> / <code>--private</code>             | whether visitors may look without signing in                                    |
| <code>--lock-roots</code>                                         | confine the folder picker to the library folders                                |
| <code>--admin-host 0.0.0.0</code>                                 | open the console to the network from the start                                  |
| <code>--strict-port</code>                                        | refuse to start rather than move to another port                                |
| <code>--name</code> , <code>--no-mdns</code>                      | the name on the home network ( <code>ninaivu.local</code> ), or none            |
| <code>--map-tiles</code>                                          | OpenStreetMap tiles on the map (they show roughly where photographs were taken) |

When the port is taken by another program, Ninaivu moves to the next free one and says so; `--port` chooses another, `--strict-port` refuses to move.

And the maintenance commands, which work in an installed copy too (the installers put a `ninaivu` command on the computer):

| Command                                                                                  |                                                                                                                                       |
|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <code>ninaivu backup --out &lt;folder&gt;</code>                                         | a copy of the index, settings and certificate                                                                                         |
| <code>ninaivu list-backups &lt;folder&gt;</code>                                         | the copies in a folder                                                                                                                |
| <code>ninaivu restore &lt;file&gt;</code>                                                | put one back (Ninaivu stopped)                                                                                                        |
| <code>ninaivu reroot --from &lt;old&gt; --to &lt;new&gt;</code>                          | tell the index the library moved                                                                                                      |
| <code>ninaivu reset-password &lt;name&gt;</code>                                         | a new password for someone, from this computer (a forgotten administrator password)                                                   |
| <code>ninaivu import-lite &lt;file&gt;</code>                                            | bring a Ninaivu Lite household across: people, who sees what, favourites, albums, share links (Ninaivu stopped, after its first scan) |
| <code>ninaivu offsite-restore --recovery &lt;file&gt; &lt;copy&gt; &lt;output&gt;</code> | put the photographs back from an off-site copy, on any computer                                                                       |

## Environment variables and the state folder

Some settings also read an environment variable ( `NINAIVU_STATE_DIR` , `NINAIVU_AI_ENGINE` , `NINAIVU_HARDWARE_TIER` , `NINAIVU_FACES` , ...; the full list is in `ninaivu/server/config.py` ). Folders and ports from the environment always apply; anything about behaviour only fills in what nobody chose in the console. The state directory ( `~/.ninaivu` , or `$XDG_DATA_HOME/ninaivu` , or `NINAIVU_STATE_DIR` ) holds the index, the thumbnails and `config.json` ; the AI models live in `.ai-models` beside the application, or `ai_models_dir` .

## Further reading

- Operations and production — Docker, systemd, reverse proxies, large libraries, backups.
- The tray, the Control Panel and HTTPS
- Backup and recovery and encryption and the recovery file
- Moving to another machine
- Contributing and security